



Project Report -3

Wazuh SIEM Attack
Simulation and
Detection



Prepared By:
ABDULLAH-AL-KAYES

Objective : Simulate brute force, port scans, and file tampering attacks from Kali Linux and detect them in Wazuh using custom rules.

Date : 06/14/2026

01 Environment

- Host : VMware Workstation
- OS : Ubuntu 24.04 LTS
- Wazuh-Version: 4.14 All-in-One
- Attacker: Kali Linux VM (same network)



02. STEP-BY-STEP DEPLOYMENT

1. Lab Network Setup :

- Deployed Kali Linux attacker VM.
- Verified ping and SSH connectivity to Ubuntu server.

2. Attack Simulations :

- **Brute Force SSH:** Ran Hydra with fake username (`dameon`) and `passwords.txt`:

```
`` `bash hydra -l dameon -P passwords.txt -t 2  
ssh://192.168.0.112 -v
```

- **Port Scanning:** Used Nmap (`nmap -sS -p 1-1000 192.168.0.112`).
- **File Tampering:** Modified files in FIM-monitored directories.

2. Custom Rule Creation :

- Added custom rule in `/var/ossec/etc/rules/local_rules.xml` (rule id = 5716).
- Restarted Wazuh manager.

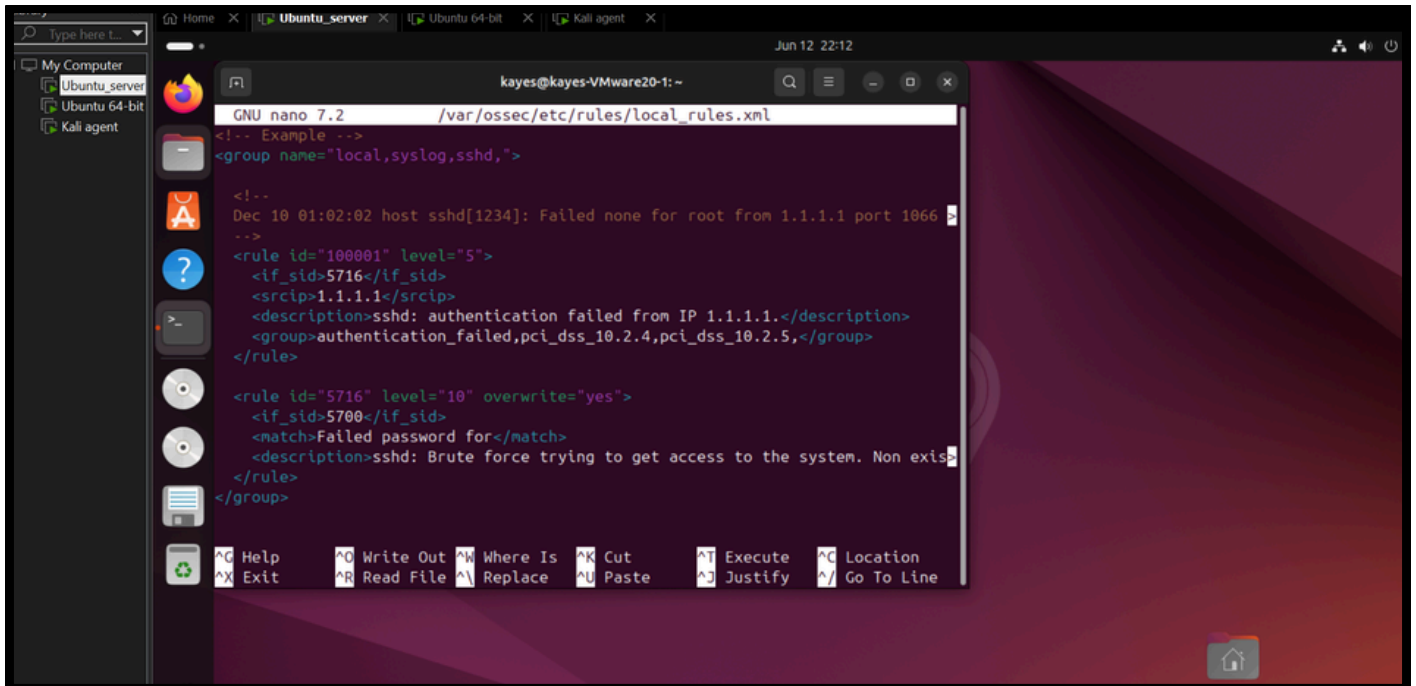
3. Alert Triage :

- Analyzed alerts in Security Events and MITRE ATT&CK tabs (Rule IDs 5710, 5716, 5763, 2502, etc.).

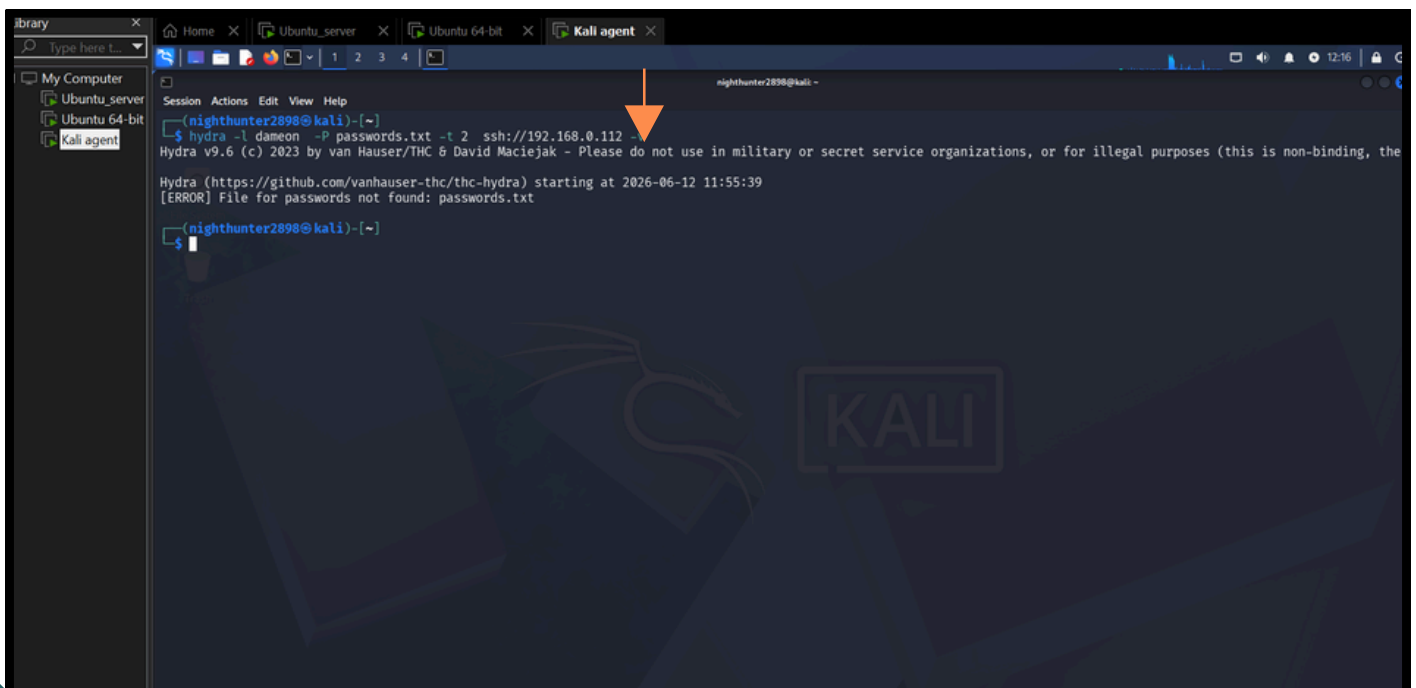


02 Screenshots

Screenshot 01 : Custom Rule in local_rules.xml

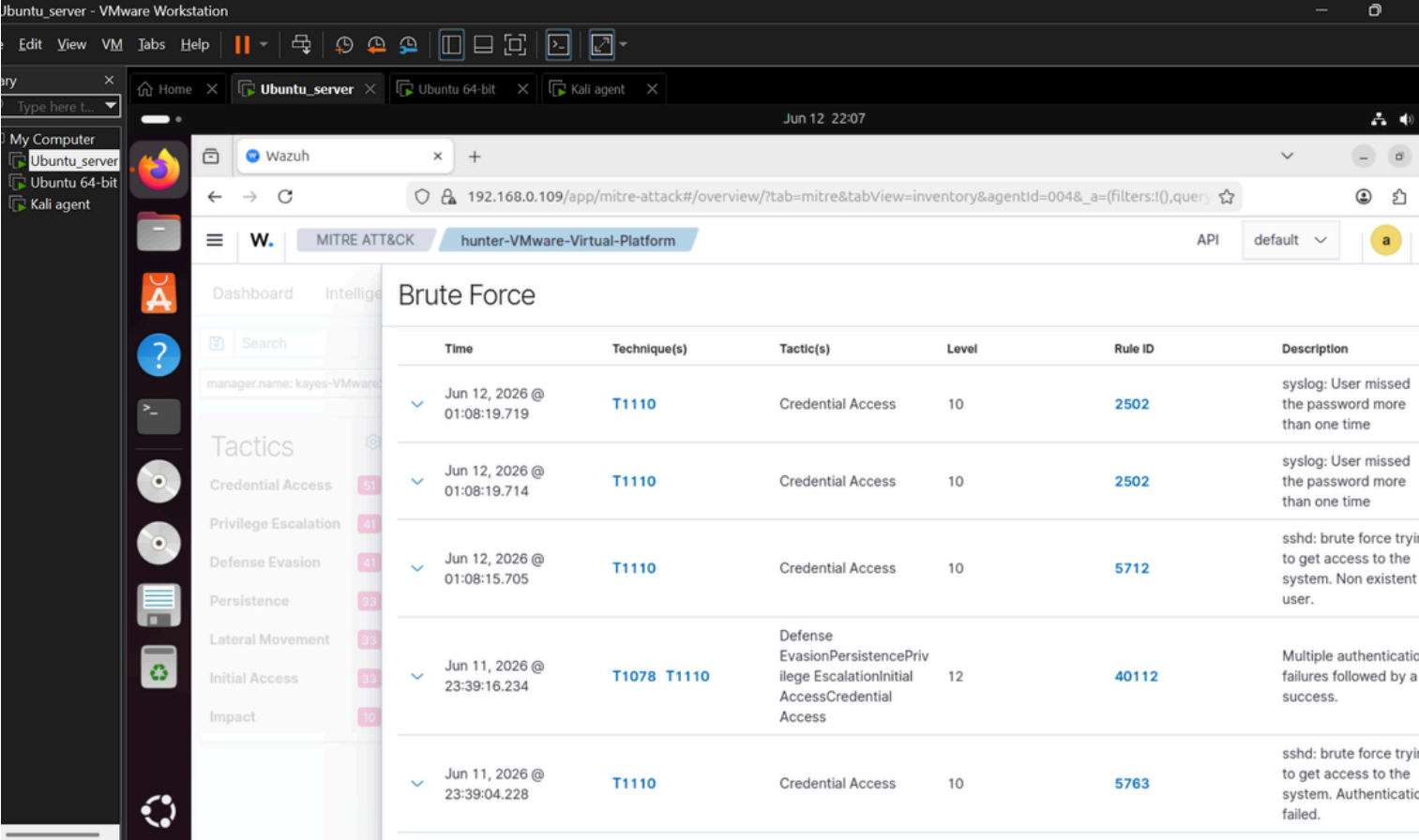


Screenshot 02 : Hydra Brute Force Attack from Kali



Screenshots

Screenshot 03: Wazuh Security Events - Brute Force Alerts

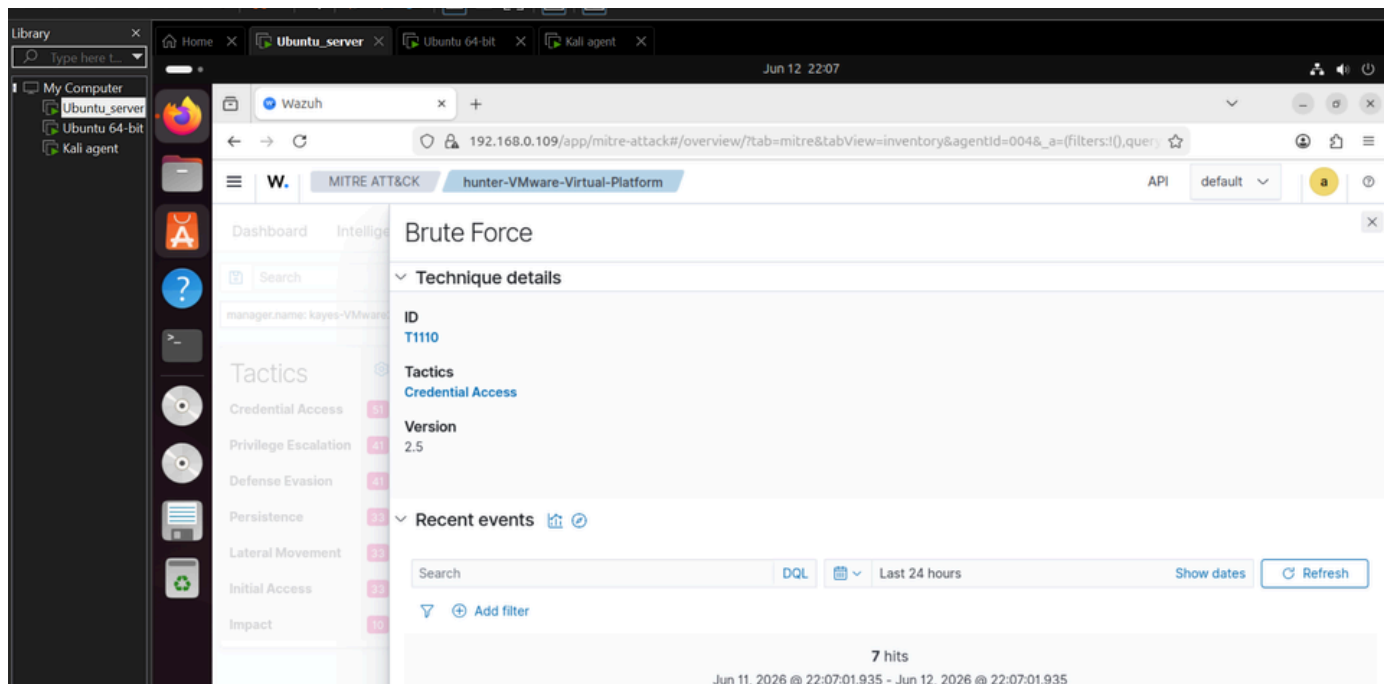


The screenshot displays the Wazuh web interface within a VMware Workstation environment. The browser window shows the 'Brute Force' alert page for the 'hunter-VMware-Virtual-Platform' agent. The page lists several security events, including password misses and failed SSH attempts.

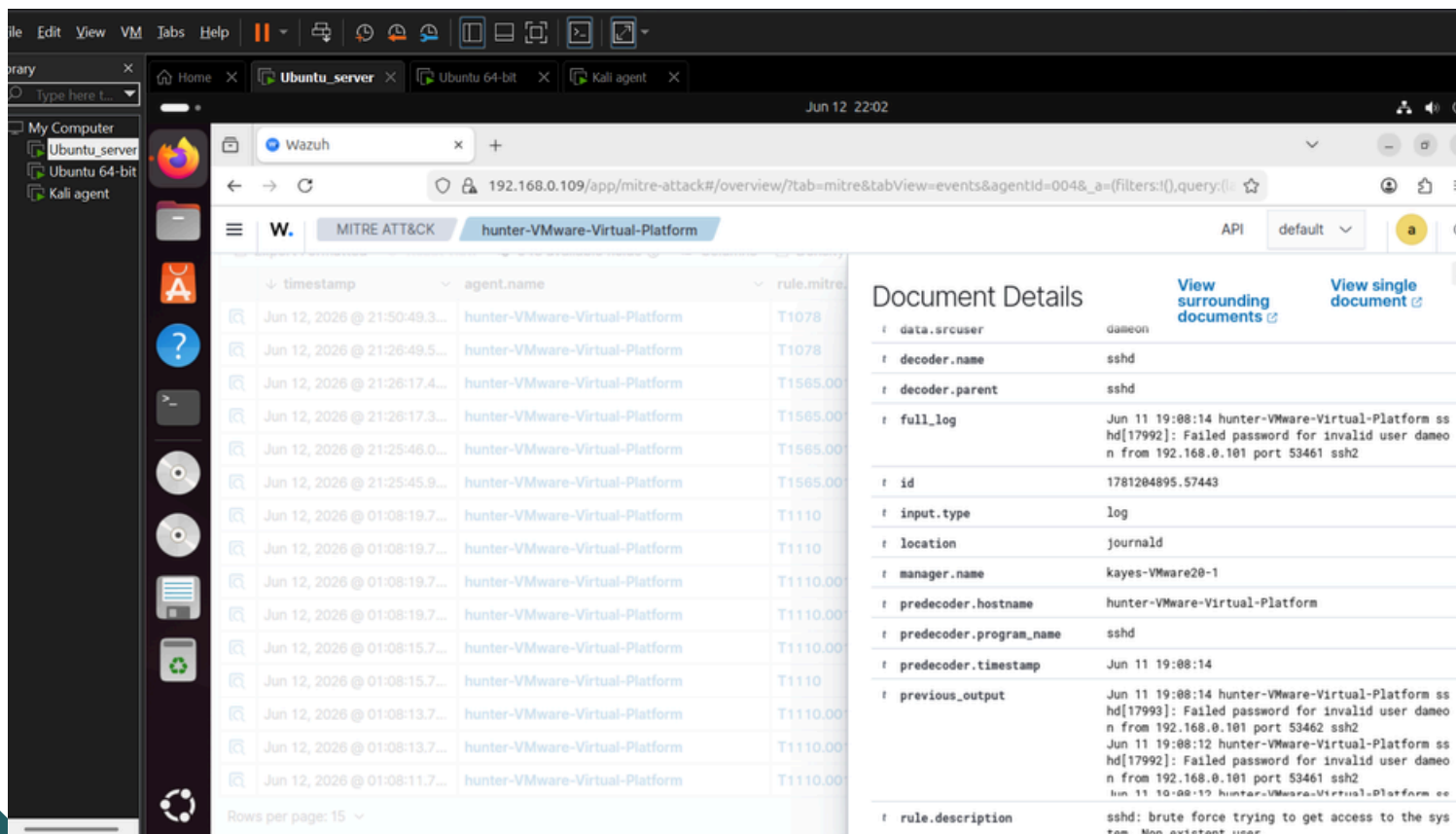
Time	Technique(s)	Tactic(s)	Level	Rule ID	Description
Jun 12, 2026 @ 01:08:19.719	T1110	Credential Access	10	2502	syslog: User missed the password more than one time
Jun 12, 2026 @ 01:08:19.714	T1110	Credential Access	10	2502	syslog: User missed the password more than one time
Jun 12, 2026 @ 01:08:15.705	T1110	Credential Access	10	5712	sshd: brute force tryir to get access to the system. Non existent user.
Jun 11, 2026 @ 23:39:16.234	T1078 T1110	Defense EvasionPersistencePrivilege EscalationInitial AccessCredential Access	12	40112	Multiple authenticatio failures followed by a success.
Jun 11, 2026 @ 23:39:04.228	T1110	Credential Access	10	5763	sshd: brute force tryir to get access to the system. Authenticatio failed.

Screenshots

Screenshot 04 : MITRE ATT&CK View (T1110)



Screenshot 05 : Alert Document Details



Screenshots

Screenshot 06 : Alert Triage Summary

The screenshot displays the Wazuh alert triage summary interface. The browser address bar shows the URL: `192.168.0.109/app/mitre-attack#/overview/?tab=mitre&tabView=inventory&agentId=004&a=(filters:[]&query=)`. The page title is "MITRE ATT&CK hunter-VMware-Virtual-Platform". The left sidebar shows the "Tactics" menu with "Credential Access" selected. The main content area displays a table of alerts under the "Brute Force" heading.

Time	Technique(s)	Tactic(s)	Level	Rule ID	Description
Jun 12, 2026 @ 01:08:19.719	T1110	Credential Access	10	2502	syslog: User missed the password more than one time
Jun 12, 2026 @ 01:08:19.714	T1110	Credential Access	10	2502	syslog: User missed the password more than one time
Jun 12, 2026 @ 01:08:15.705	T1110	Credential Access	10	5712	sshd: brute force trying to get access to the system. Non existent user.
Jun 11, 2026 @ 23:39:16.234	T1078 T1110	Defense Evasion Persistence Privilege Escalation Initial Access Credential Access	12	40112	Multiple authentication failures followed by a success.
Jun 11, 2026 @ 23:39:04.228	T1110	Credential Access	10	5763	sshd: brute force trying to get access to the system. Authentication failed.

Screenshot 06 : Nmap Scan Execution & SSH.

The screenshot shows a terminal window on a Kali agent. The terminal displays the output of an Nmap scan and an SSH connection attempt. The user is logged in as "nighthunter2898@kali".

```
nighthunter2898@kali:~$ nmap -p 1-1000 192.168.0.112
Nmap scan report for 192.168.0.112
Host is up (0.0027s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 10.01 seconds

nighthunter2898@kali:~$ ssh 192.168.0.112
Warning: Permanently added '192.168.0.112' (ssh) to the list of known hosts.
nighthunter2898@kali:~$
```

03 Challenges & Solutions

- Challenge : Frequent IP changes causing agent disconnection.
Solution: Plan to configure static IP on Ubuntu server.
- Challenge: Hydra file/path errors.
Solution: Created passwords.txt in ~/Documents. This way hydra could be executed without any issues.
- Challenge : Default rules missing some brute force events.
Solution : Added custom rule and used -t 2 for controlled testing. For controlled slow attacks so that the SIEM can detect them properly.

05. SOC Skills Gained

- Attack simulation with Hydra and Nmap
- Custom rule development in Wazuh
- Brute force detection (MITRE T1110)
- Alert triage and investigation
- Red/Blue team exercise workflow